



■ CS177 Bioinformatics: Software Development and Applications

Lecture 11: Selected Topics of AI

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Outline



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- Generative Adversarial Networks (GANs)
- Diffusion models
- Knowledge Graphs (KGs)
- Explainable AI (XAI)



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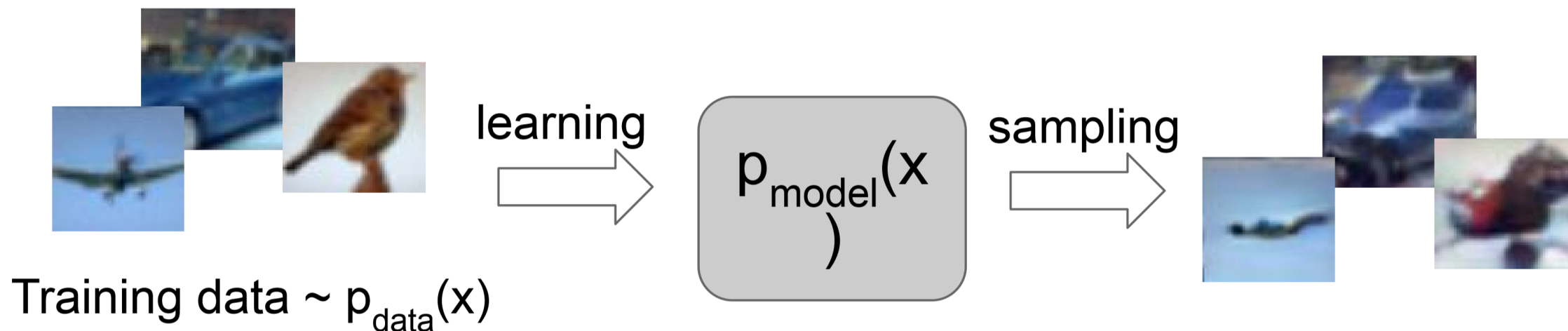
Generative Adversarial Networks (GANs)



Generative modeling



- Given training data, generate new samples from the same distribution

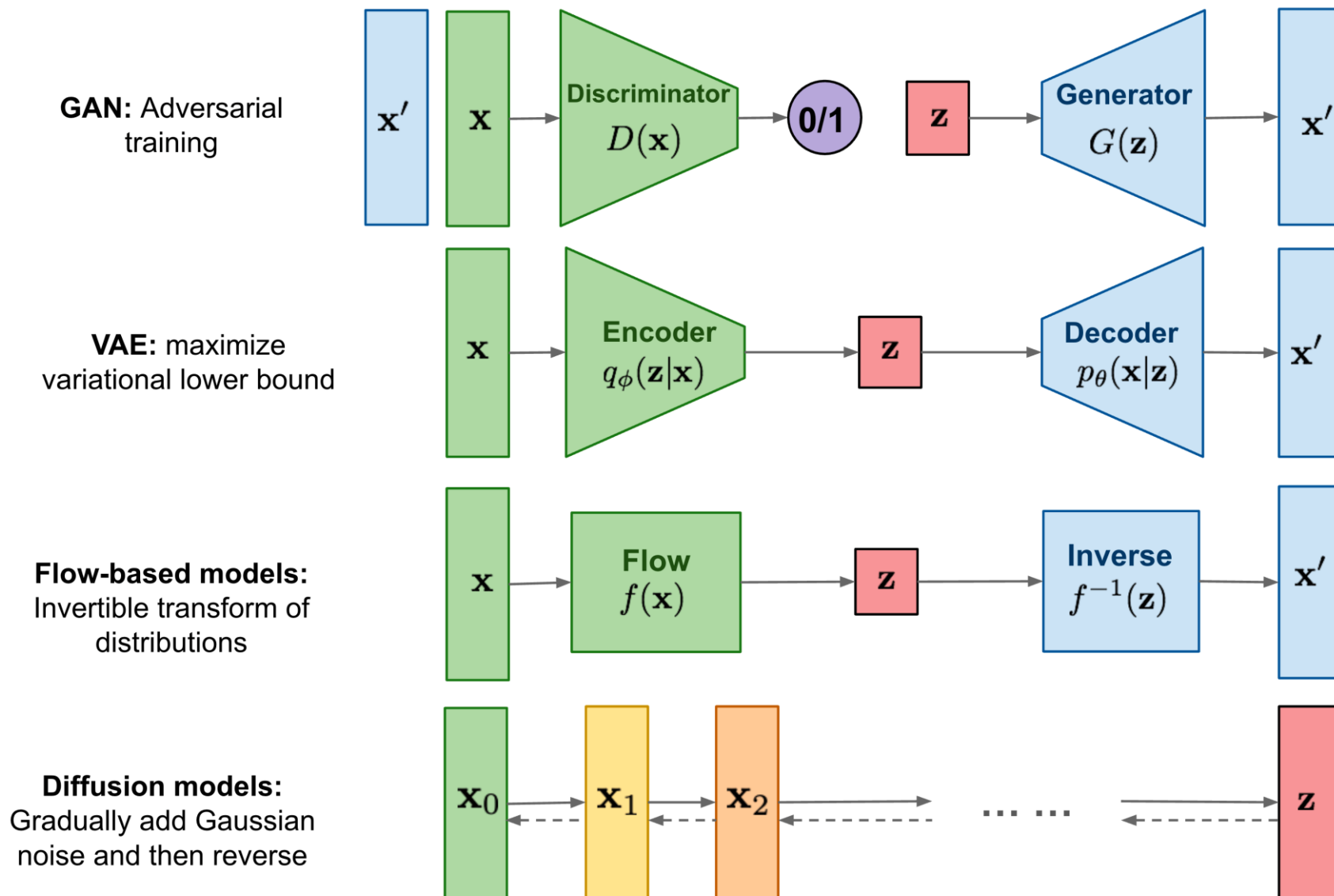


Objectives:

1. Learn $p_{\text{model}}(x)$ that approximates $p_{\text{data}}(x)$
2. Sample new x from $p_{\text{model}}(x)$



Generative models

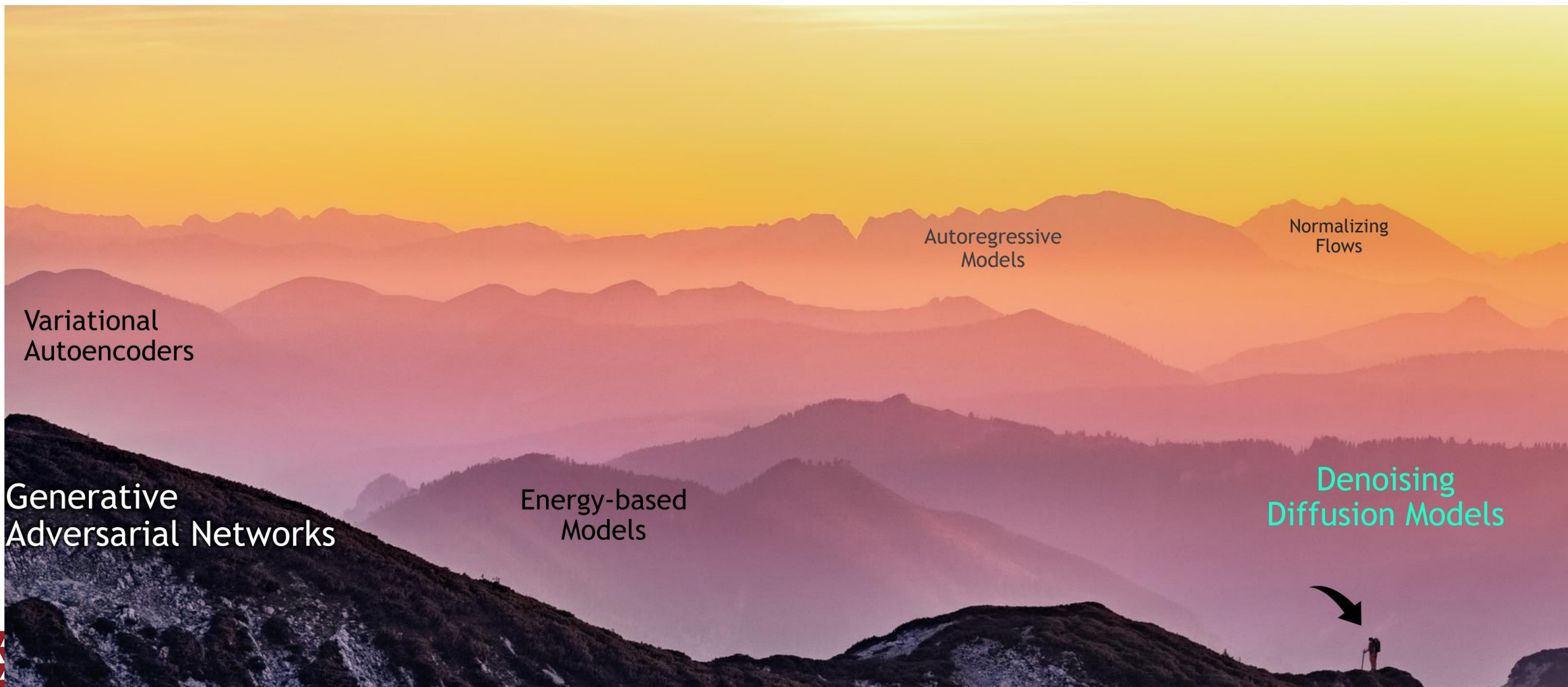




The landscape of deep generative learning



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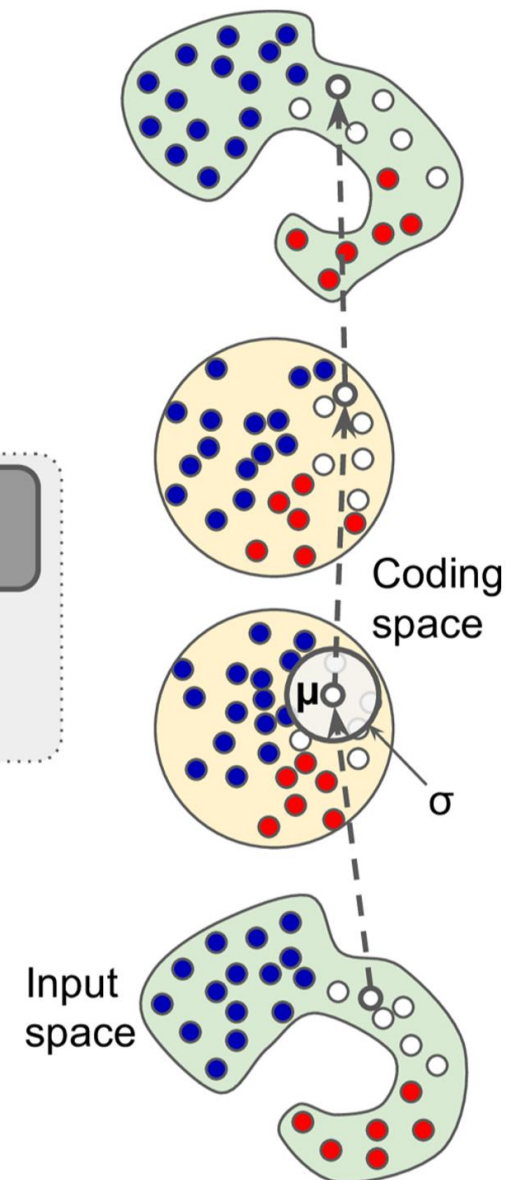
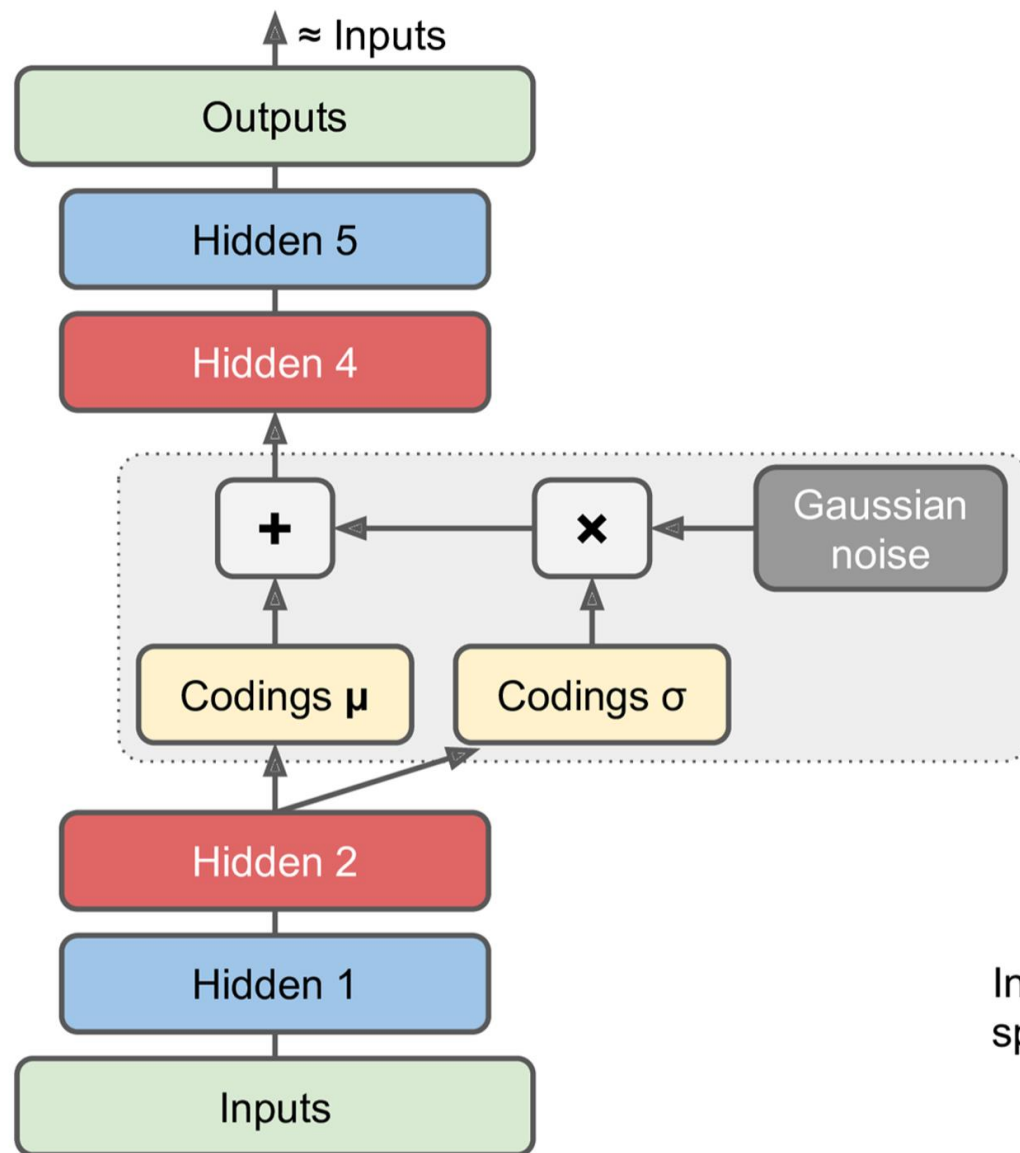
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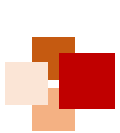


Recap: Variational Autoencoder (VAE)



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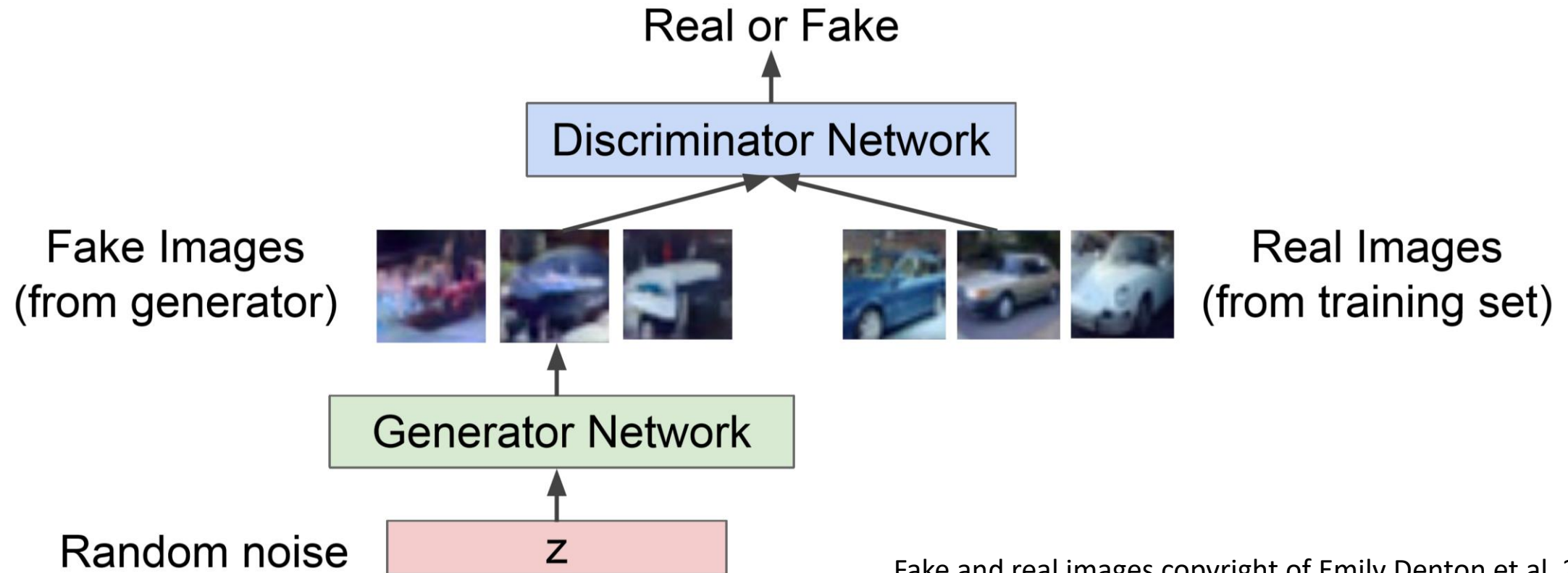


Generative Adversarial Networks (GANs)



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- Use a neural network, called **generator**, to generate data
- Use another neural network, called **discriminator**, to determine if the data is real or fake



Fake and real images copyright of Emily Denton et al. 2015.



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Cost functions of adversarial learning



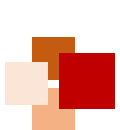
- Let D denote the discriminator's predicted probability of being real data
- **Discriminator:**
 - Try to distinguish between real and fake images
 - **Cost function:** cross-entropy loss for the task of classifying real vs. fake images:

$$\mathcal{J}_D = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[-\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}}[-\log(1 - D(G(\mathbf{z})))]$$

- **Generator:**
 - Try to fool the discriminator by generating real-looking images
 - **Cost function** (one possible version) is the opposite of the discriminator's:

$$\begin{aligned}\mathcal{J}_G &= -\mathcal{J}_D \\ &= \text{const} + \mathbb{E}_{\mathbf{z}}[\log(1 - D(G(\mathbf{z})))]\end{aligned}$$





Two-player game



- Minimax formulation:
 - The generator and discriminator are playing a **zero-sum game** against each other
 - Train jointly in a **minimax game**:

Discriminator outputs likelihood in (0,1) of real image

Minimax objective function:

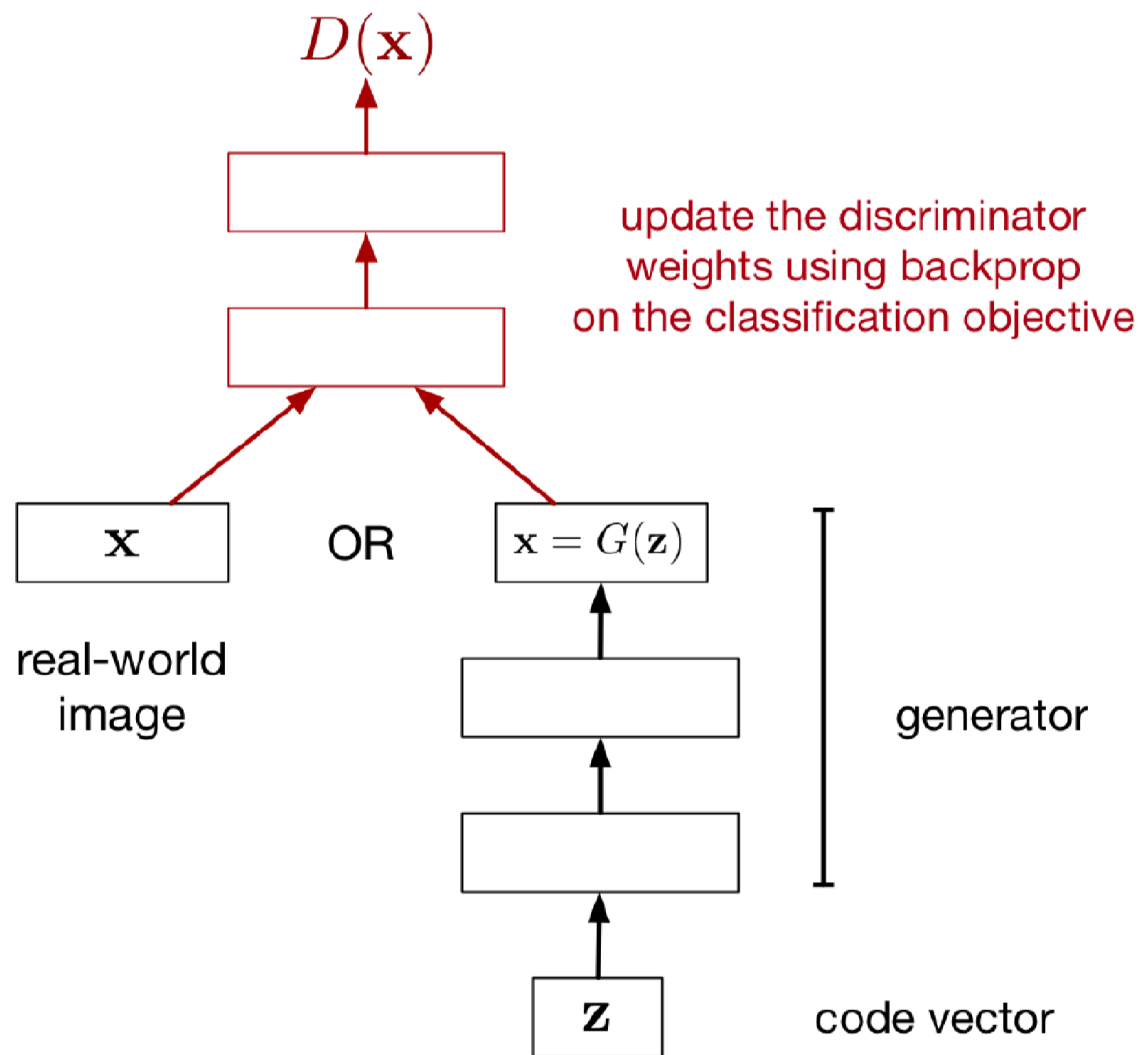
$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log \underbrace{D_{\theta_d}(x)}_{\substack{\text{Discriminator output} \\ \text{for real data } x}} + \mathbb{E}_{z \sim p(z)} \log(1 - \underbrace{D_{\theta_d}(G_{\theta_g}(z))}_{\substack{\text{Discriminator output for} \\ \text{generated fake data } G(z)}}) \right]$$



Learning procedure



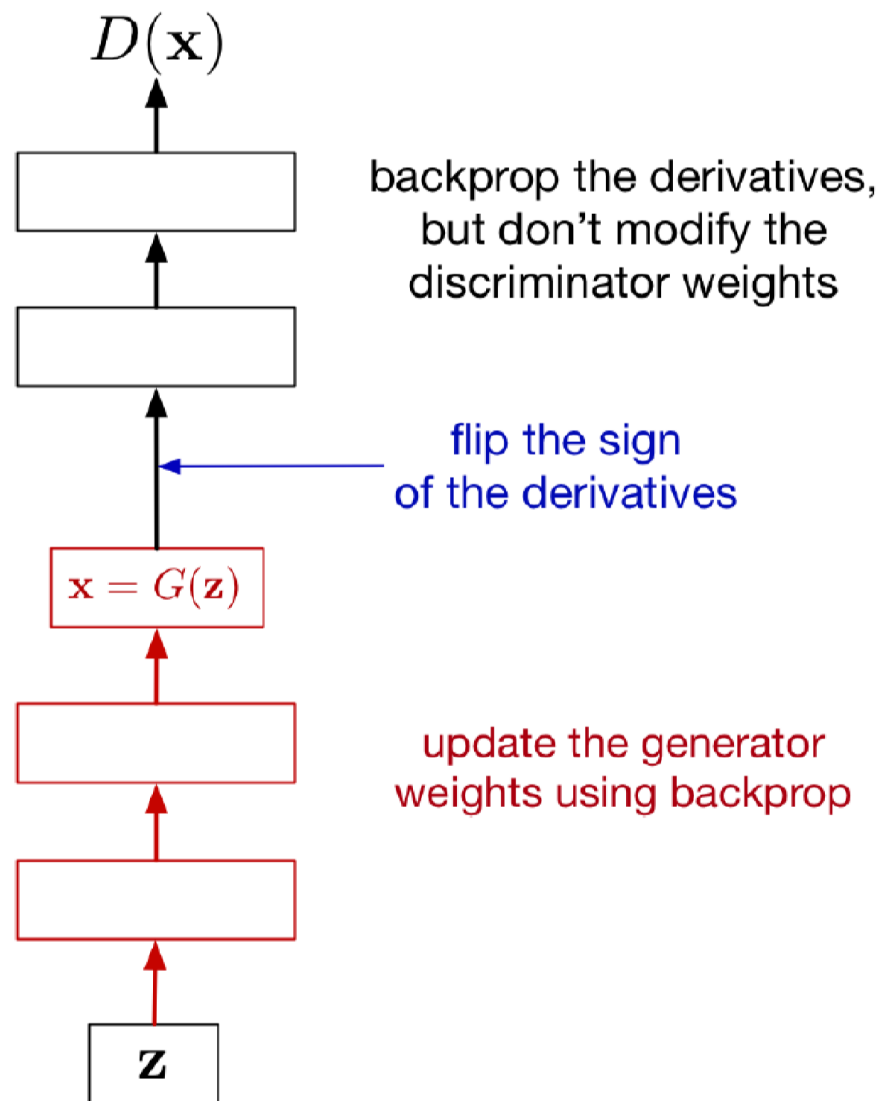
- Updating the discriminator



Learning procedure

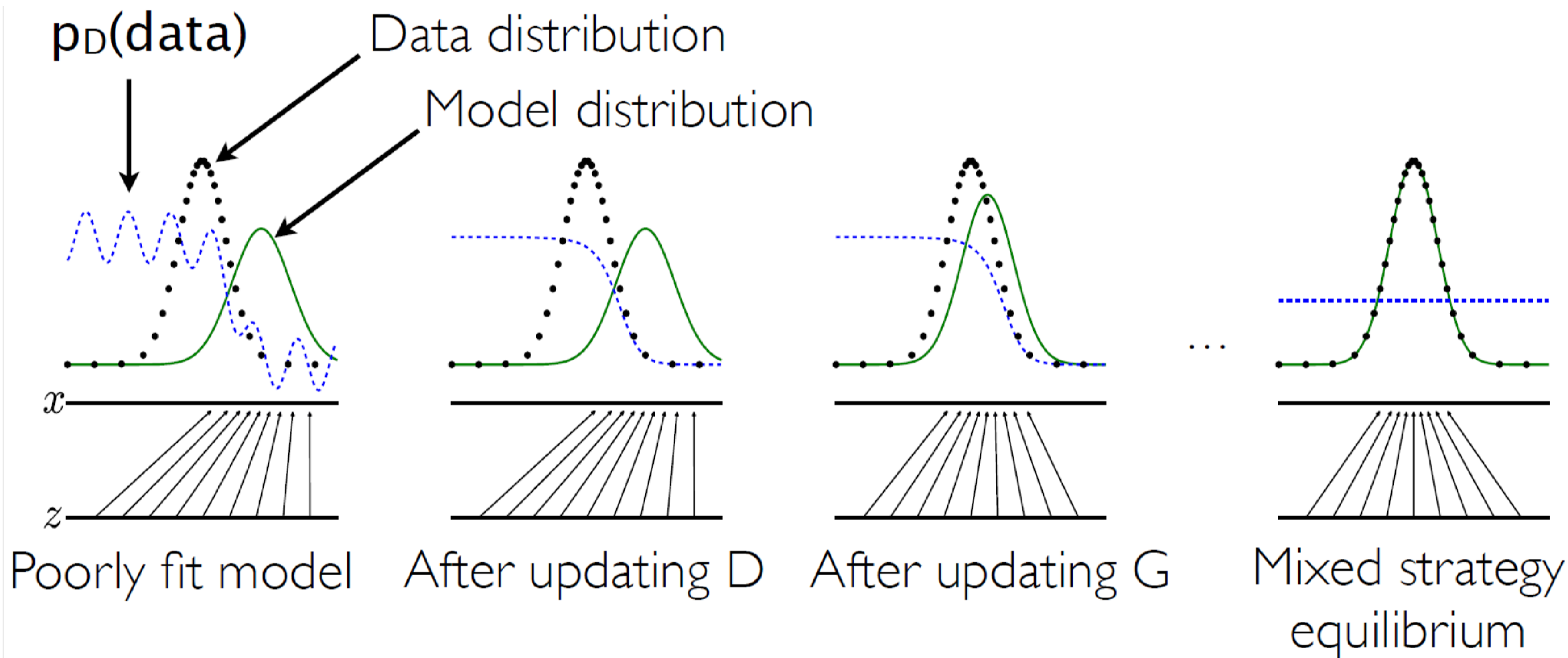


- Updating the generator





Training GANs



Modified from Fig. 1 of Ian J. Goodfellow et al. NIPS, 2014.





Training GANs



- In general, training a GAN is tricky and unstable
- Since GANs were introduced in 2014, there have been hundreds of papers introducing various architectures and training methods
- GAN Zoo: <https://github.com/hindupuravinash/the-gan-zoo>
- Many tricks:
 - S. Chintala, How to train a GAN, ICCV 2017 tutorial
 - <https://github.com/soumith/ganhacks>





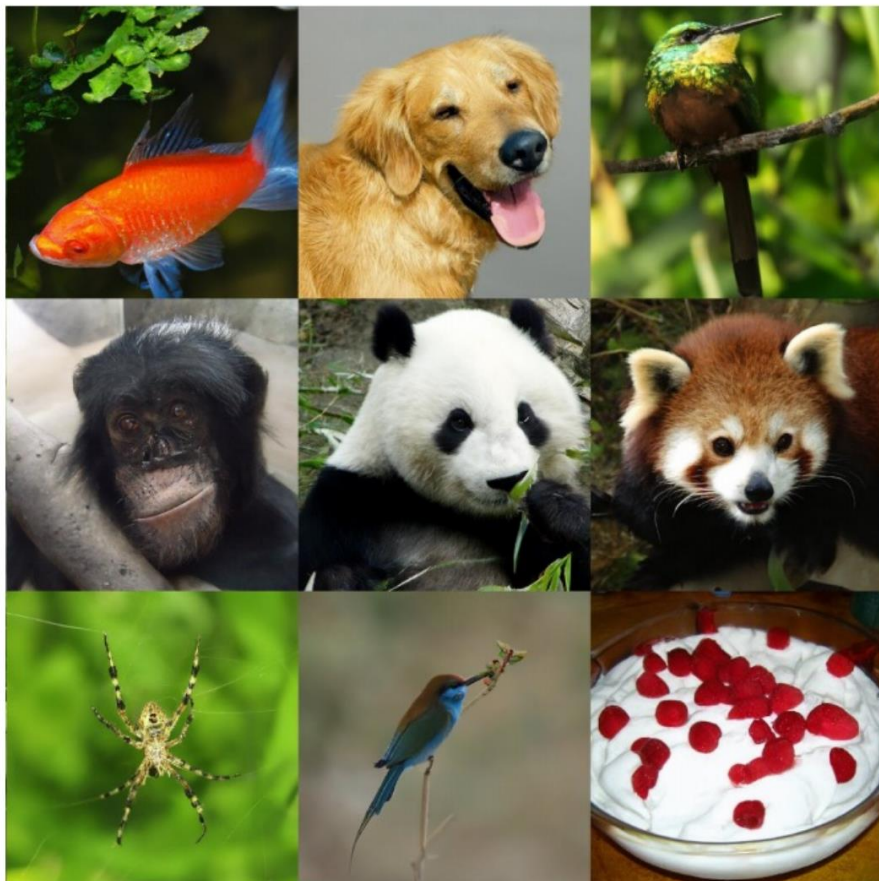
Diffusion Models

PPT from CVPR 2022 tutorial “Denoising Diffusion-based Generative Modeling: Foundations and Applications” by K. Kreis, R. Gao and A. Vahdat



Denoising Diffusion Models

Emerging as powerful generative models, outperforming GANs



[“Diffusion Models Beat GANs on Image Synthesis”](#)
Dhariwal & Nichol, OpenAI, 2021



[“Cascaded Diffusion Models for High Fidelity Image Generation”](#)
Ho et al., Google, 2021

Text-to-Image Generation

DALL·E 2

“a teddy bear on a skateboard in times square”



[“Hierarchical Text-Conditional Image Generation with CLIP Latents”](#)
Ramesh et al., 2022

Imagen

A group of teddy bears in suit in a corporate office celebrating the birthday of their friend. There is a pizza cake on the desk.



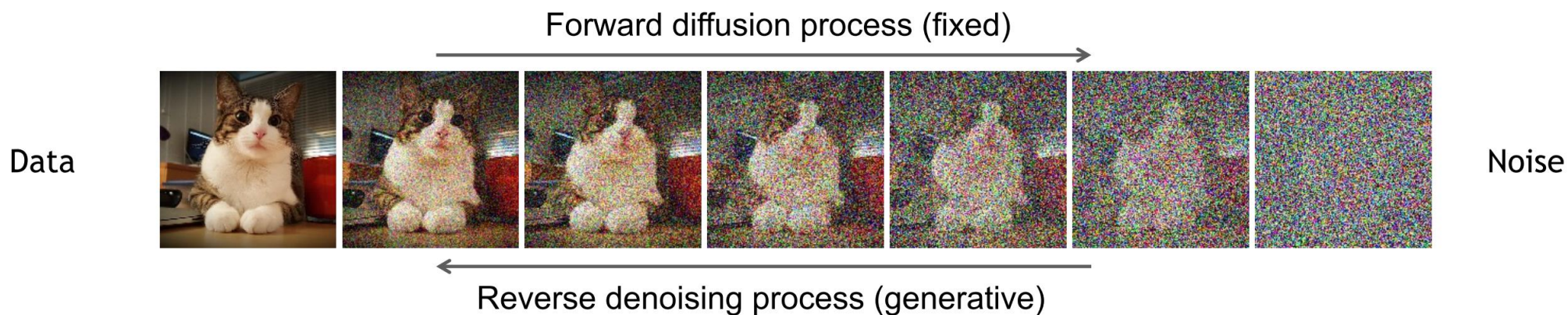
[“Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding”](#), Saharia et al., 2022

Denoising Diffusion Models

Learning to generate by denoising

Denoising diffusion models consist of two processes:

- Forward diffusion process that gradually adds noise to input
- Reverse denoising process that learns to generate data by denoising



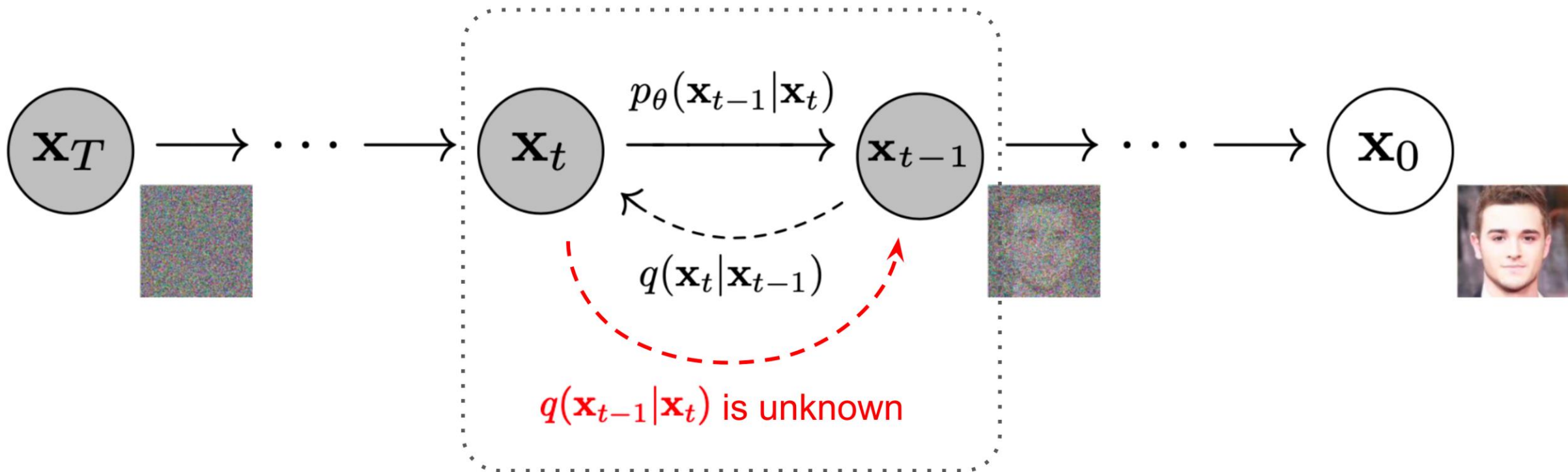
[Sohl-Dickstein et al., Deep Unsupervised Learning using Nonequilibrium Thermodynamics, ICML 2015](#)

[Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020](#)

[Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021](#)

■ Denoising Diffusion Probabilistic Models (DDPMs)

Use variational lower bound



Ho, Jonathan, Ajay Jain, and Pieter Abbeel. NeurIPS 2020.

Conditional diffusion models

Include condition as input to reverse process

Reverse process: $p_{\theta}(\mathbf{x}_{0:T}|\mathbf{c}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c}), \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t, \mathbf{c}), \Sigma_{\theta}(\mathbf{x}_t, t, \mathbf{c}))$

Variational upper bound: $L_{\theta}(\mathbf{x}_0|\mathbf{c}) = \mathbb{E}_q \left[L_T(\mathbf{x}_0) + \sum_{t>1} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c})) - \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1, \mathbf{c}) \right].$

Incorporate conditions into U-Net

- Scalar conditioning: encode scalar as a vector embedding, simple spatial addition or adaptive group normalization layers.
- Image conditioning: channel-wise concatenation of the conditional image.
- Text conditioning: single vector embedding - spatial addition or adaptive group norm / a seq of vector embeddings - cross-attention.

Imagen

Google Research, Brain team

Input: text; Output: 1kx1k images

- An unprecedented degree of photorealism
 - SOTA automatic scores & human ratings
- A deep level of language understanding
- Extremely simple
 - no latent space, no quantization

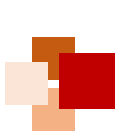


A brain riding a rocketship heading towards the moon.



Knowledge Graph (KG)





Background



- Incorporating human knowledge is an important direction of AI
- Knowledge representation and reasoning, inspired by human problem solving, can enable AI systems to solve complex tasks (e.g. medical diagnosis)
- Recently, knowledge graphs as a form of structured human knowledge have drawn great attention from both academia and industry
- Knowledge representation has a long history in logic and AI

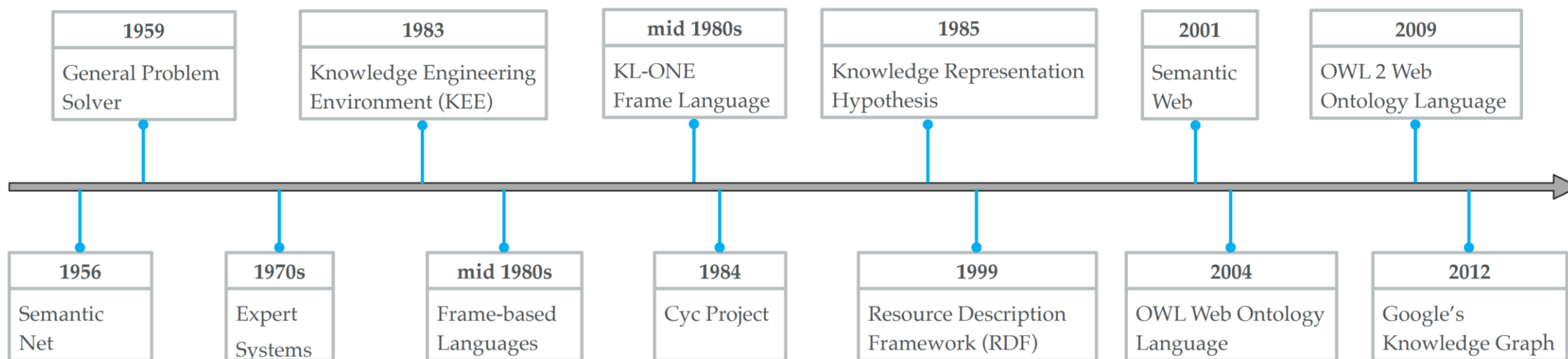
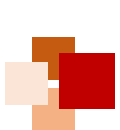


Figure source: Ji et al. A survey on knowledge graphs: representation, acquisition and applications, IEEE Trans. Neural Networks and Learning Systems, 2021



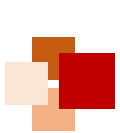


Definition of knowledge graph (KG)



- There is no wide-accepted formal definition as yet.
- **Definition 1** (Ehrlinger and Wöβ, 2016): A **knowledge graph** acquires and integrates information into an ontology and applies a reasoner to derive new knowledge.
 - **Ontology**: A set of concepts and categories in a subject area or domain that shows their properties and the relations between them.
- **Definition 2** (Wang et al. 2017): A **knowledge graph** is a multi-relational graph composed of entities and relations which are regarded as nodes and different types of edges, respectively.
- **Definition 3** (Ji et al. 2021): A **knowledge graph** is defined as $G = \{E, R, F\}$, where E is the set of **entities**, R is the set of **relations**, and F is the set of **facts**, respectively.
 - A **fact** is denoted as a triple $(h, r, t) \in F$.



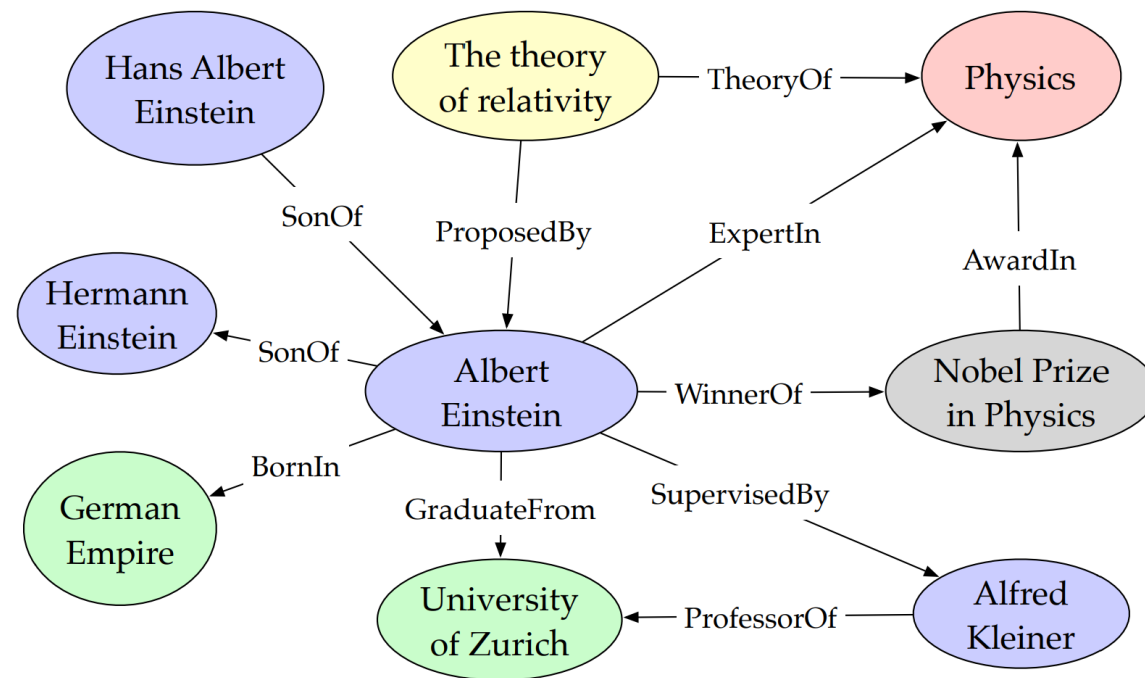


An example of knowledge base and KG



(Albert Einstein, **BornIn**, German Empire)
(Albert Einstein, **SonOf**, Hermann Einstein)
(Albert Einstein, **GraduateFrom**, University of Zurich)
(Albert Einstein, **WinnerOf**, Nobel Prize in Physics)
(Albert Einstein, **ExpertIn**, Physics)
(Nobel Prize in Physics, **AwardIn**, Physics)
(The theory of relativity, **TheoryOf**, Physics)
(Albert Einstein, **SupervisedBy**, Alfred Kleiner)
(Alfred Kleiner, **ProfessorOf**, University of Zurich)
(The theory of relativity, **ProposedBy**, Albert Einstein)
(Hans Albert Einstein, **SonOf**, Albert Einstein)

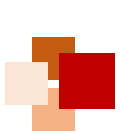
(a) Factual triples in a knowledge base.



(b) Entities and relations in a knowledge graph.

Figure source: Ji et al. A survey on knowledge graphs: representation, acquisition and applications, IEEE Trans. Neural Networks and Learning Systems, 2021





A roadmap of research on KG

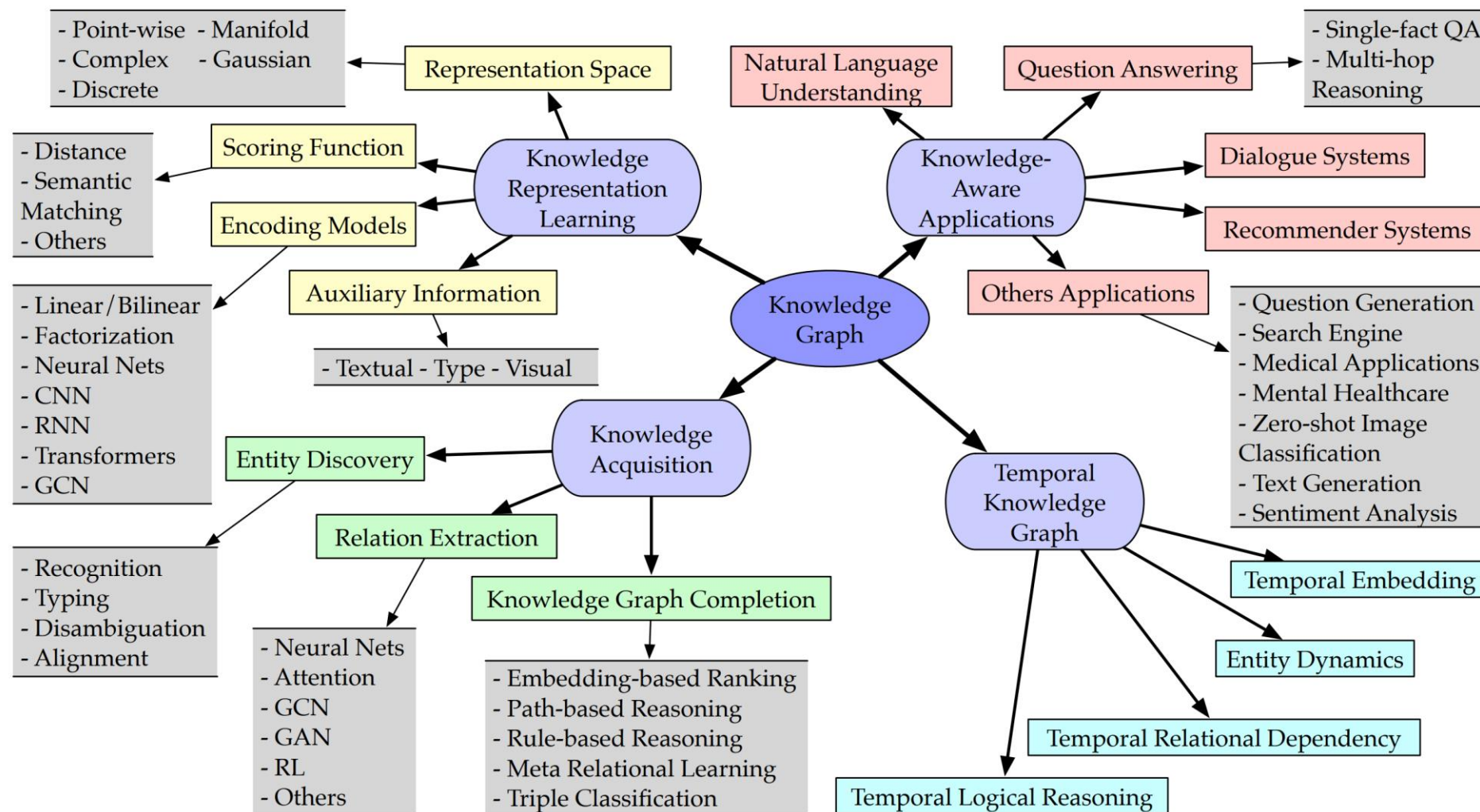
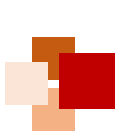


Figure source: Ji et al. A survey on knowledge graphs: representation, acquisition and applications, IEEE Trans. Neural Networks and Learning Systems, 2021

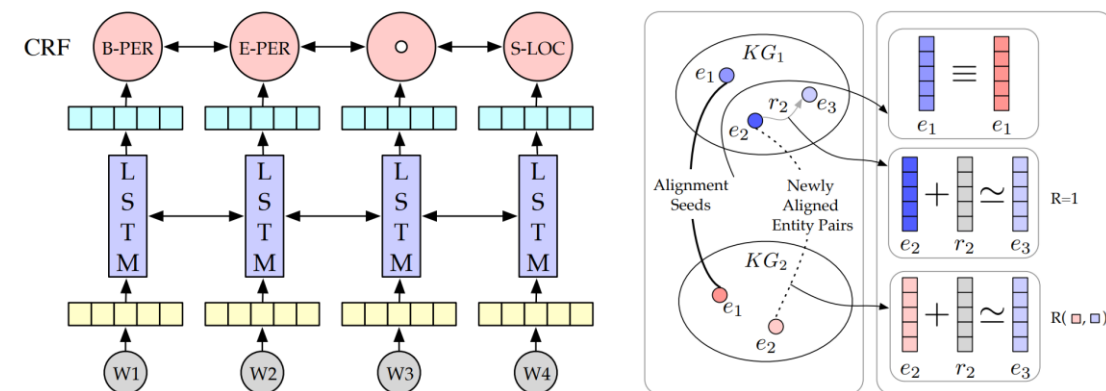




Building KG: Knowledge Acquisition



- **Knowledge graph completion** completes missing links between existing entities or infers entities given entity and relation queries
- **Entity discovery** acquires entity-oriented knowledge from text and fuses knowledge between knowledge graphs
- **Relation extraction** is to build large-scale KGs automatically by extracting unknown relational facts from plain text and adding them into KGs



(a) Entity recognition with LSTM-CRF [111]. (b) Entity alignment with IP-TransE [126].

Fig. 8: Illustrations of several entity discovery tasks.

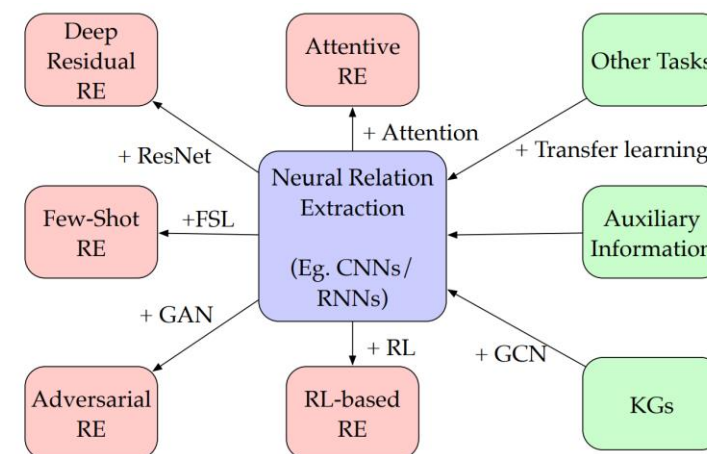


Fig. 9: An overview of neural relation extraction.

Figure source: Ji et al. IEEE Trans. Neural Networks and Learning Systems, 2021



Knowledge-aware applications



- **Challenge:** How to integrate symbolic knowledge into computational frameworks of real-world applications?
- Applications of KGs include two folds:
 - 1) **In-KG**: link prediction, named entity recognition (NER)
 - 2) **Out-of-KG**: relation extraction, more downstream applications such as question answering and recommender systems
- Example applications:
 - Language representation learning
 - Question answering
 - Recommender systems
 - Text classification
 - Dialogue systems
 - Medicine and Biology



Application in DrugBank



Ritonavir

Summary Ritonavir is an HIV protease inhibitor used in combination with other antivirals in the treatment of HIV infection.

Brand Names Kaletra, Norvir, Viekira Pak

Potential Drug Targets

Show 10 entries

DRUG	TARGET
Ritonavir	Human immunodeficiency virus type 1 protease
Darunavir	Human immunodeficiency virus type 1 protease
Lopinavir	Human immunodeficiency virus type 1 protease
Methylprednisolone	Glucocorticoid receptor
Leronlimab	C-C chemokine receptor type 5
Tocilizumab	Interleukin-6 receptor subunit alpha
Colchicine	Tubulin beta chain
Galidesivir	RNA-directed RNA polymerase
Favipiravir	RNA-directed RNA polymerase
TMC-310911	HIV-1 protease

Showing 1 to 10 of 385 entries

DrugBank DB00503

Human immunodeficiency virus type 1 protease

DETAILS

Name Human immunodeficiency virus type 1 protease

Kind protein

Organism Human immunodeficiency virus 1

Protein

NAME	UNIPROT ID
Pol polyprotein	Q72874 Details

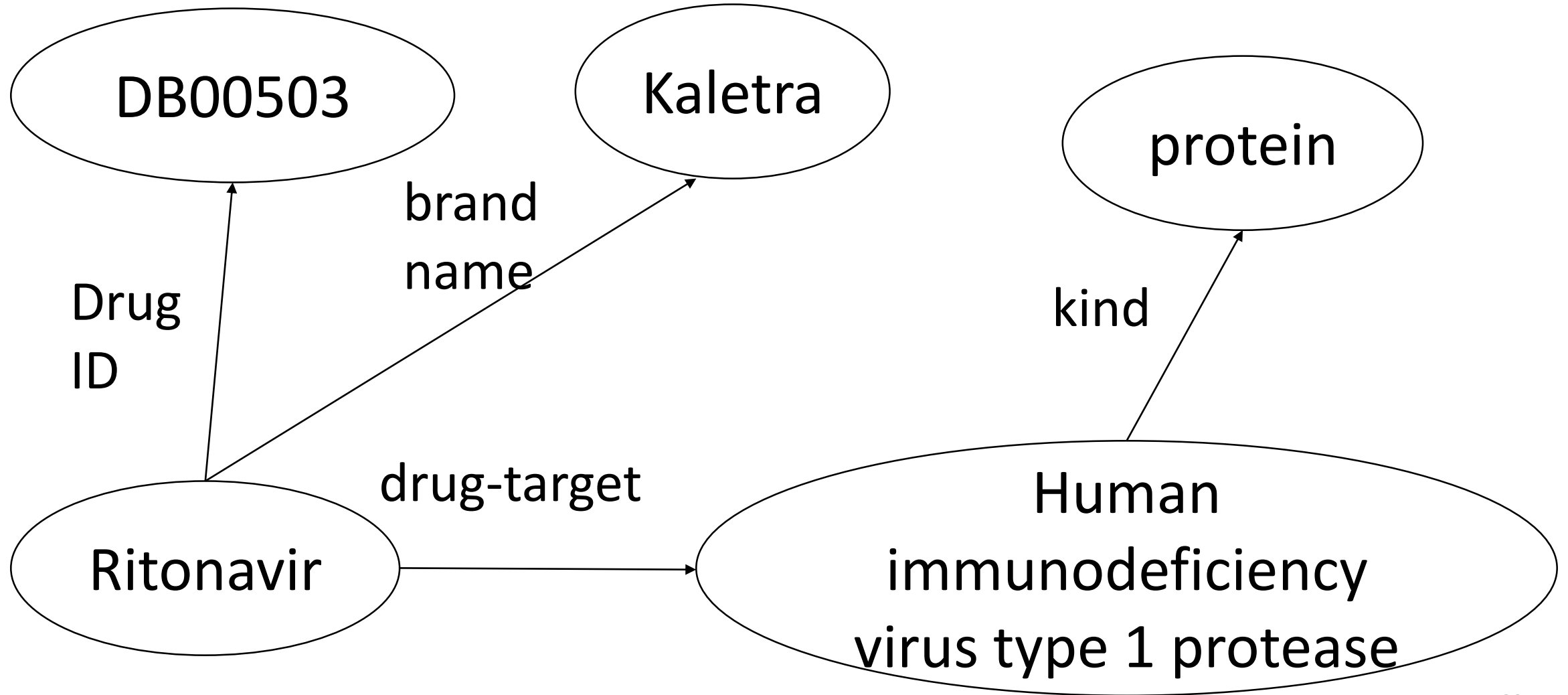
DRUG RELATIONS

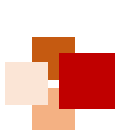
Drug Relations

Show 10 entries

DRUGBANK ID	NAME	DRUG GROUP	PHARMACOLOGICAL ACTION?
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Application in DrugBank

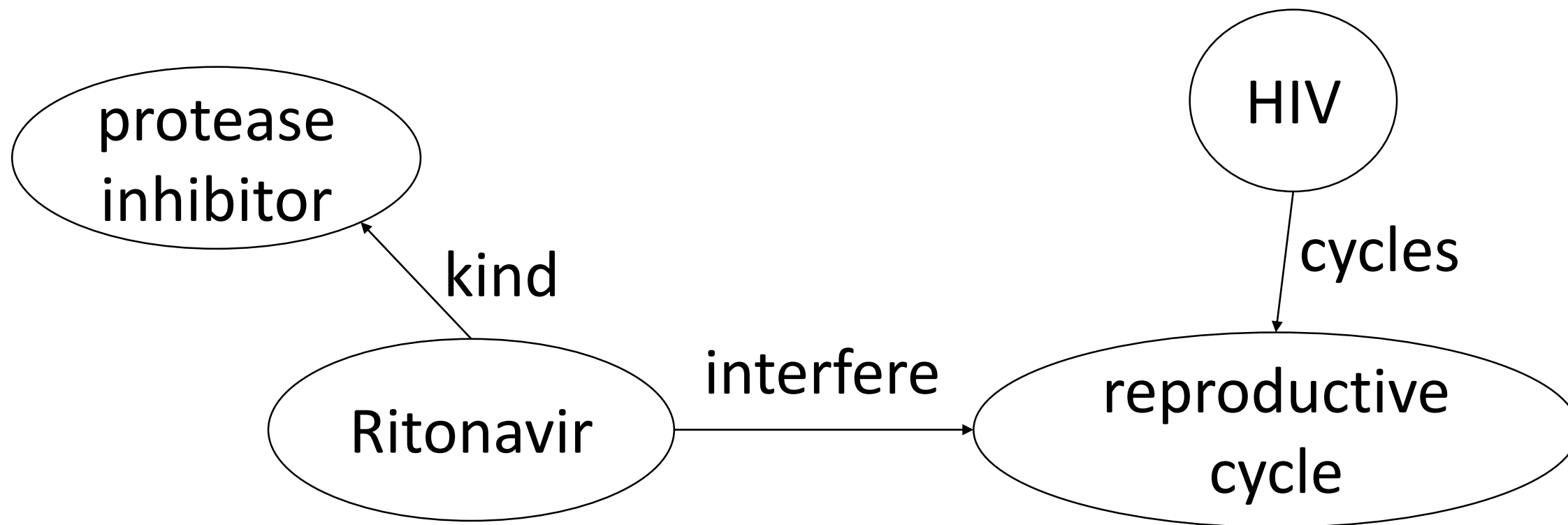


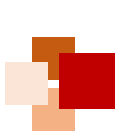


Entity Recognition & Relation Extraction

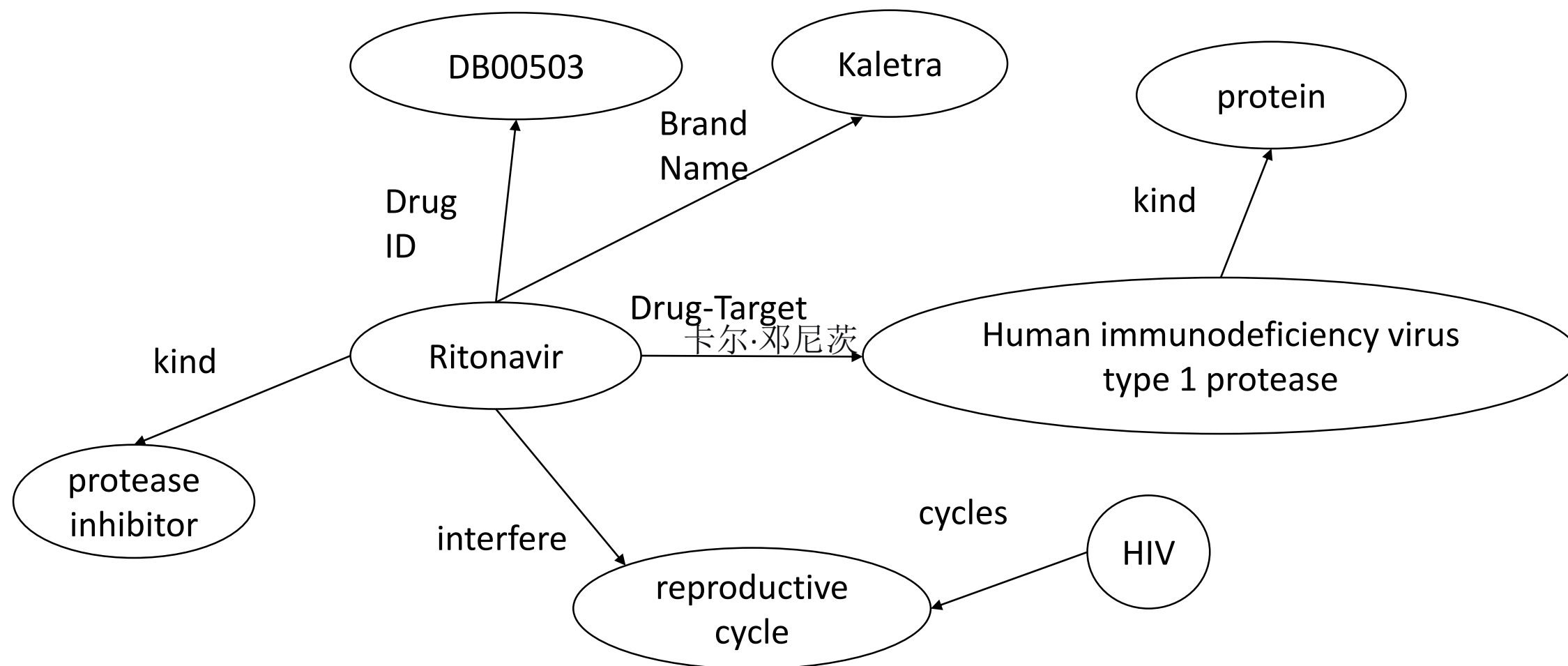


- **Ritonavir** *is* an **HIV protease inhibitor** that *interferes* with the **reproductive cycle** *of* HIV





A small KG from DrugBank





KG behind Search Engine of DrugBank



Search over 500,000 drugs & drug products on DrugBank Online

Tylenol



Drugs



Targets



Pathways



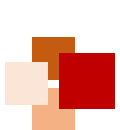
Indications





Explainable AI (XAI)





Explainable AI (XAI) - what



Explainability: associated with the notion of explanation as an interface between humans and a decision maker, that is, both an accurate proxy of the decision maker and comprehensible to humans

*Given an audience, an **explainable** Artificial Intelligence is one that produces details or reasons to make its **functioning clear or easy to understand.***

Arrieta et al. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 2020, 58: 82-115.





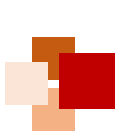
Explainable AI (XAI) - why



- To fill the gap between the research community and business sectors or users out of the domain
- Opens the door for further model improvement and its practical utility

Arrieta et al. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 2020, 58: 82-115.





The taxonomy of XAI methods



Dimension 1 — Passive vs. Active Approaches

Passive	Post hoc explain trained neural networks
Active	Actively change the network architecture or training process for better interpretability

Dimension 2 — Type of Explanations (in the order of increasing explanatory power)

To explain a prediction/class by

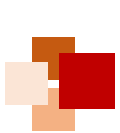
Examples	Provide example(s) which may be considered similar or as prototype(s)
Attribution	Assign credit (or blame) to the input features (e.g. feature importance, saliency masks)
Hidden semantics	Make sense of certain hidden neurons/layers
Rules	Extract logic rules (e.g. decision trees, rule sets and other rule formats)

Dimension 3 — Local vs. Global Interpretability (in terms of the input space)

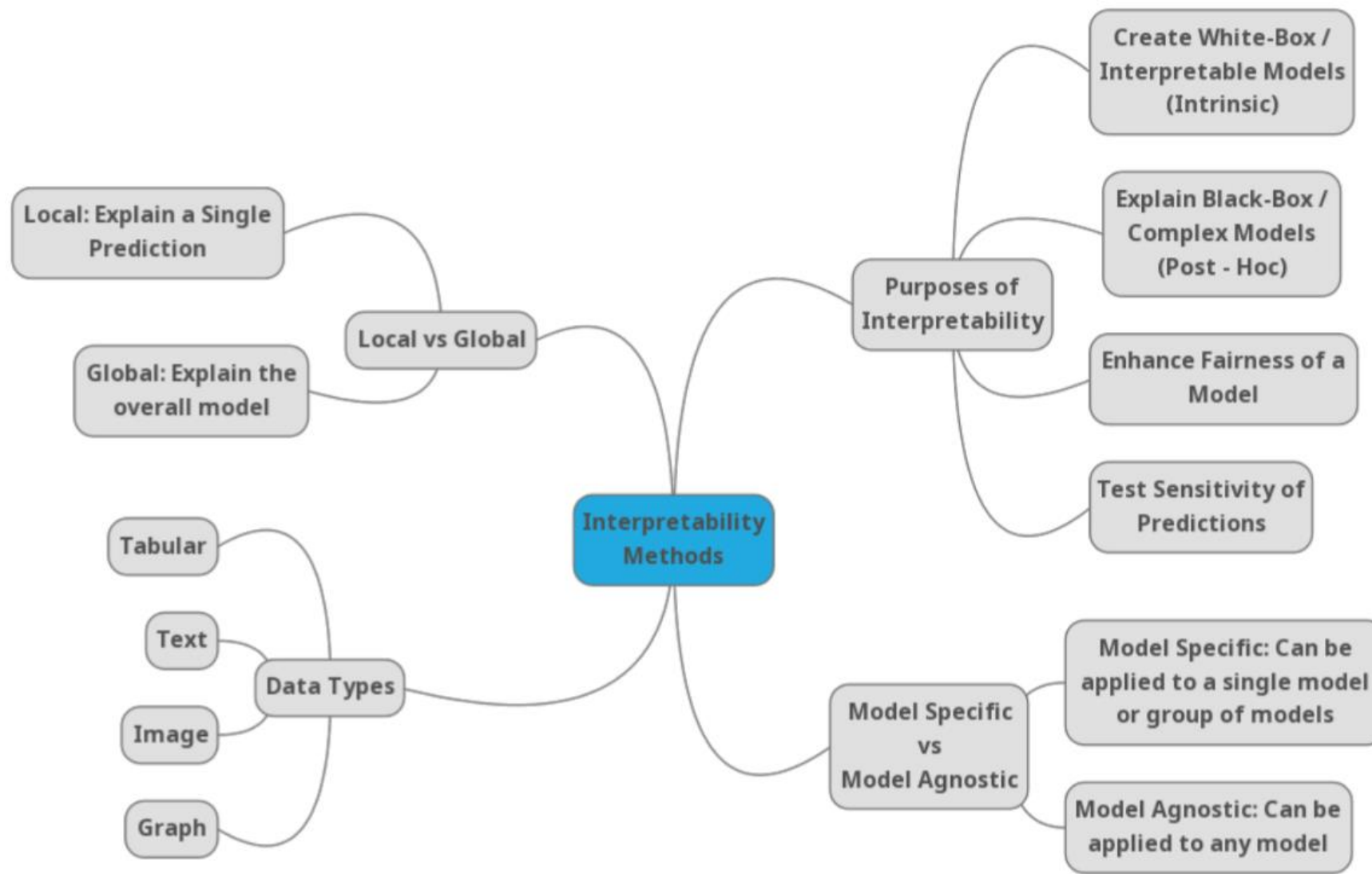
Local	Explain network's <i>predictions on individual samples</i> (e.g. a saliency mask for an input image)
Semi-local	In between, for example, explain a group of similar inputs together
Global	Explain the network <i>as a whole</i> (e.g. a set of rules/a decision tree)

Zhang et al. A survey on neural network interpretability. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2021.

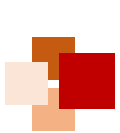




The taxonomy of XAI methods



Linardatos et al. Explainable AI: A review of machine learning interpretability methods. *Entropy*, 23(1), 18, 2021.



Overview of current XAI methods



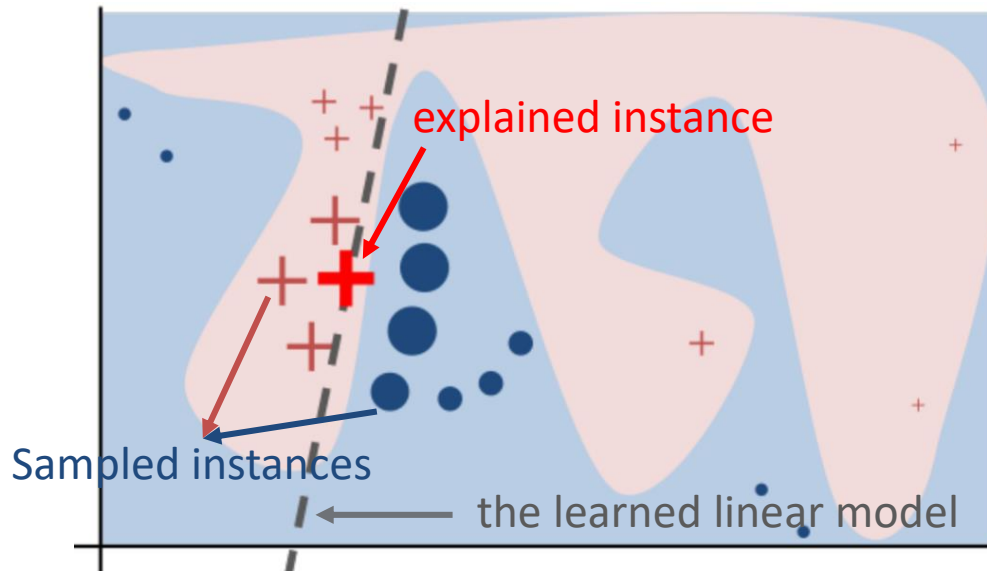
		Local	Semi-local	Global
Passive	Rule	CEM [69], CDRPs [77], CVE ² [78], DACE [79]	Anchors [65], Interpretable partial substitution [80]	KT [81], M of N [82], NeuralRule [83], NeuroLinear [84], GRG [85], Gyan ^{FO} [86], •FZ [87], [88], Trepan [89], • [90], DecText [91], Global model on CEM [92]
	Hidden semantics	(*No explicit methods but many in the below cell can be applied here.)	—	Visualization [71], [93]–[98], Network dissection [21], Net2Vec [99], Linguistic correlation analysis [100]
	Attribution ¹	LIME [20], MAPLE [101], Partial derivatives [71], DeconvNet [72], Guided backprop [102], Guided Grad-CAM [103], Shapley values [104]–[107], Sensitivity analysis [72], [108], [109], Feature selector [110], Bias attribution [111]	DeepLIFT [112], LRP [113], Integrated gradients [114], Feature selector [110], MAME [68]	Feature selector [110], TCAV [63], ACE [115], SpRAY ³ [67], MAME [68], DeepConsensus [116]
	By example	Influence functions [73], Representer point selection [117]	—	—
Active	Rule	—	Regional tree regularization [118]	Tree regularization [119]
	Hidden semantics	—	—	“One filter, one concept” [70]
	Attribution	ExpO [120], DAPr [121]	—	Dual-net (feature importance) [122]
	By example	—	—	Network with a prototype layer [76], ProtoPNet [123]



XAI example: LIME

- LIME (Local Interpretable Model-Agnostic Explanations)

Idea: sample instances to learn a linear model that could locally approximate the original predictions



Prediction probabilities

atheism	0.58
christian	0.42

atheism

christian

Posting 0.15
Host 0.14
NNTP 0.11
edu 0.04
have 0.01
There 0.01

Text with highlighted words

From: johnchad@triton.unm.edu (jchadwic)

Subject: Another request for Darwin Fish

Organization: University of New Mexico, Albuquerque

Lines: 11

NNTP-Posting-Host: triton.unm.edu

Hello Gang,

There have been some notes recently asking where to obtain the DARWIN fish.

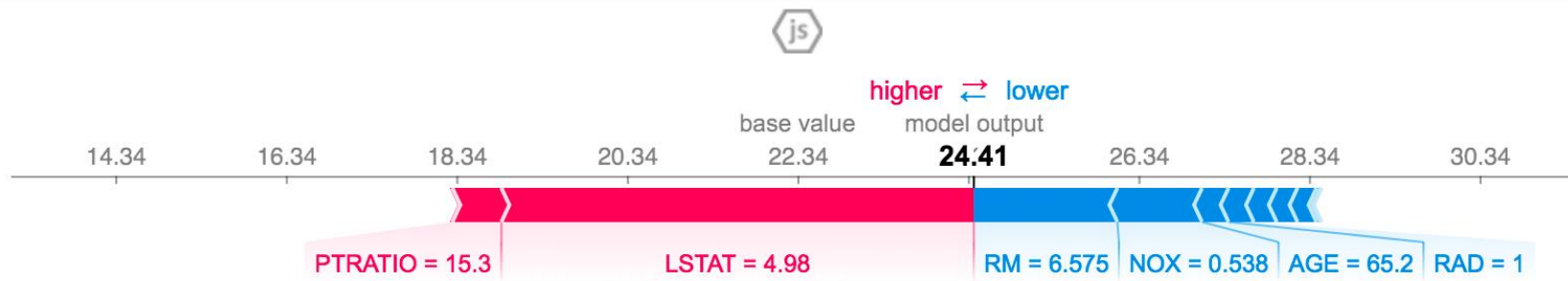
This is the same question I have and I have not seen an answer on the

net. If anyone has a contact please post on the net or email me.

Ribeiro et al. Why should I trust you? Explaining the predictions of any classifier. *KDD*, 2016.

XAI example: SHAP

- **SHAP**, or **SHapley Additive exPlanations**, is a game theoretic approach to explain the output of any machine learning model
 - It connects **optimal credit allocation** with **local explanations** using the classic **Shapley values** from cooperative game theory and their related extensions
- In machine learning, a **Shapley value** is the contribution of a feature value to difference between the actual prediction and the mean prediction
 - Introduced in 1953 by Lloyd Shapley (who won Nobel Prize in Economics, 2012), one of the fathers of Game Theory
 - Intuition: sometimes a player value in a team could be greater than their value if they were on their own. For machine learning, we can view the features as the “players” and the outcome of the model as the “game”’s result”
 - Given that without any features we would just predict an average value, once we bring the first feature in, how much our prediction would change compared to the average?

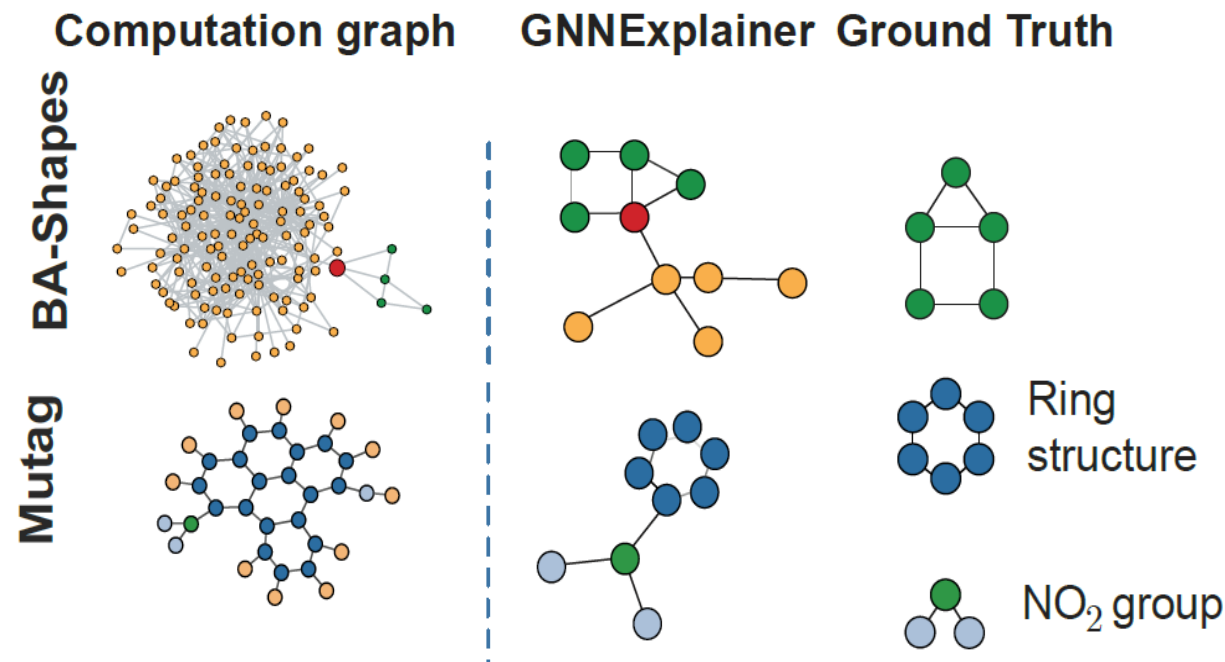
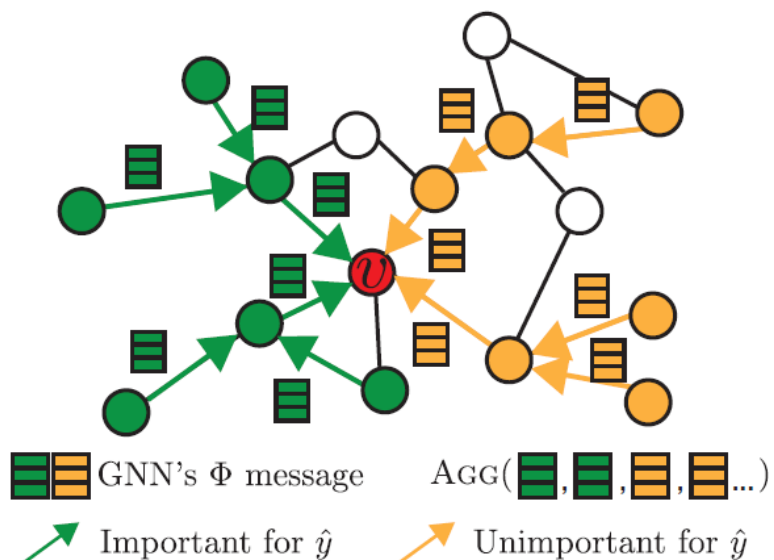


Lundberg & Lee. A unified approach to interpreting model predictions. *NIPS*, 2017.

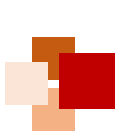
XAI example: GNNExplainer

• GNNExplainer

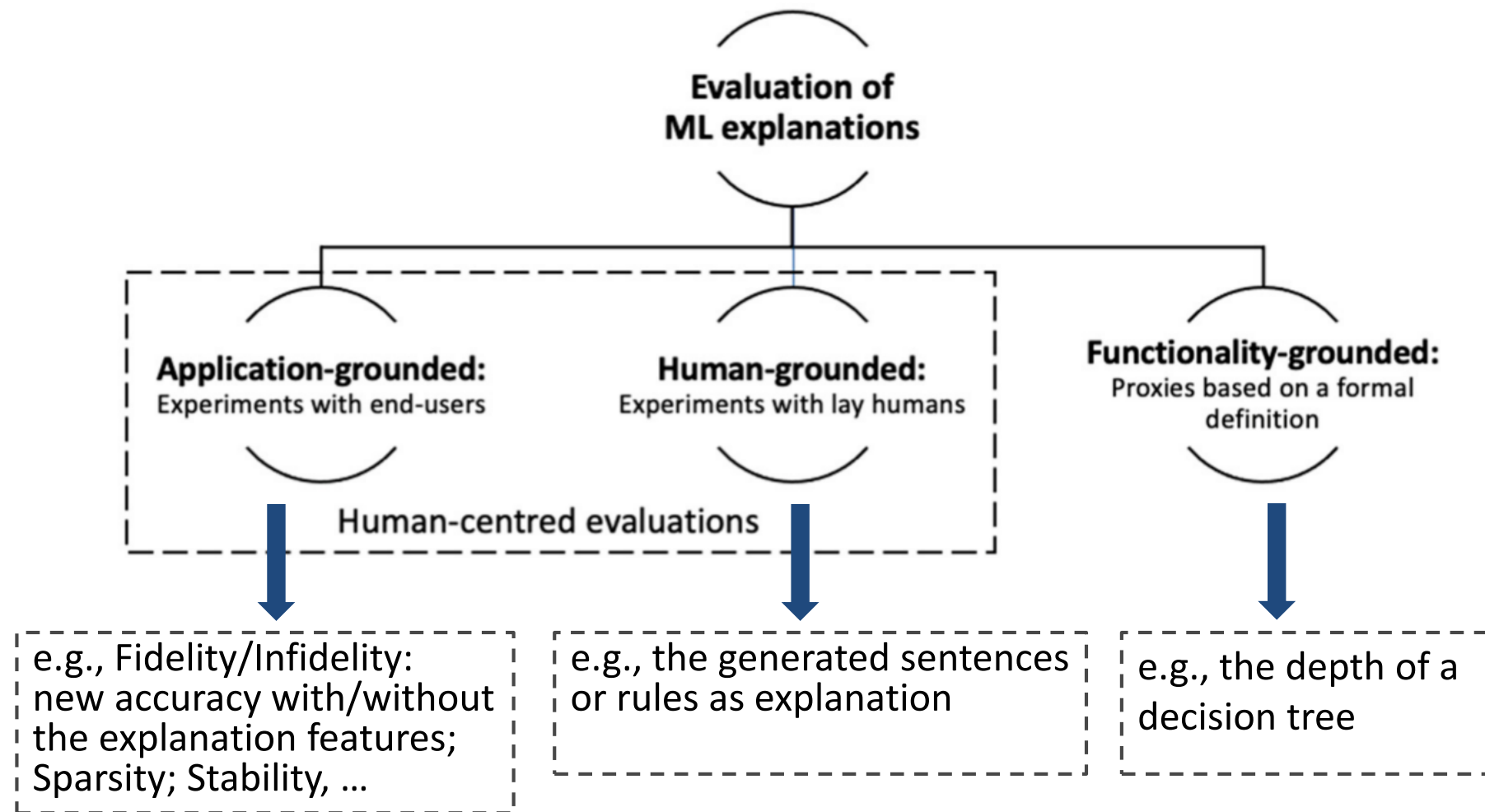
Learning a subgraph to approximate the predictions of a GNN model which uses a whole graph



Ying et al. GNNExplainer: Generating explanations for graph neural networks. *NIPS*, 2019.



Evaluating XAI methods



[1] Zhou et al. Evaluating the quality of machine learning explanations: A survey on methods and metrics. *Electronics*, 2021.

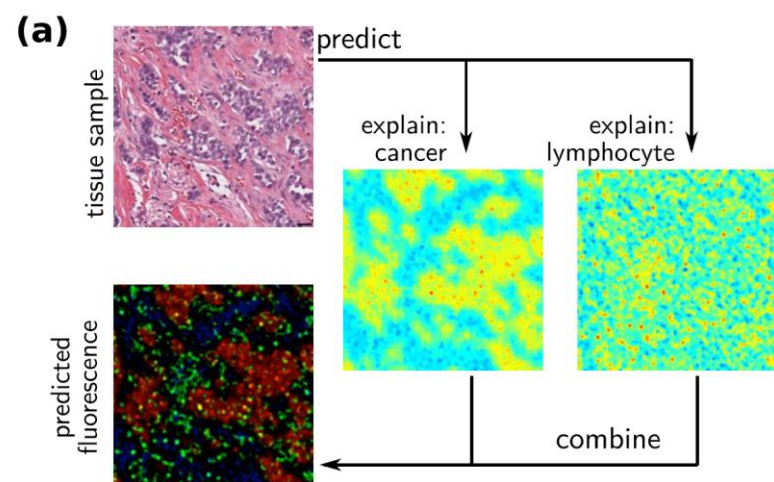
[2] Yuan et al. Explainability in graph neural networks: A taxonomic survey. *arXiv*, 2020.



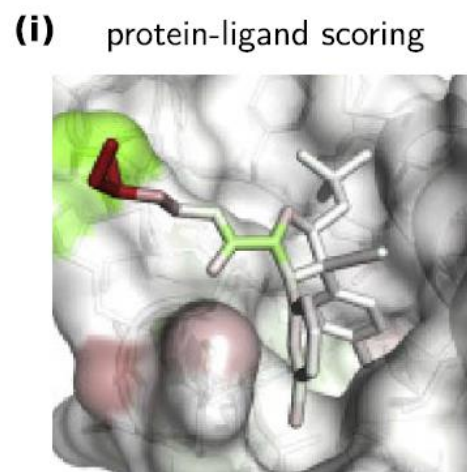
Applications of XAI



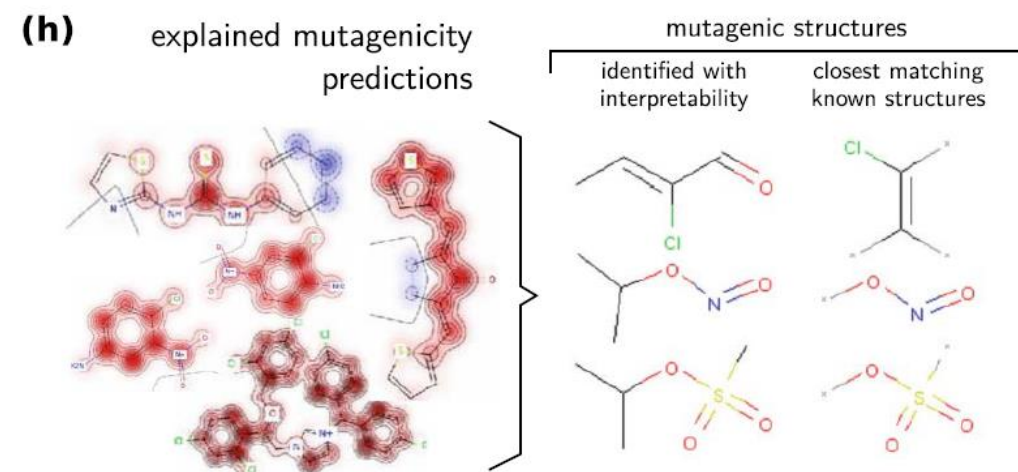
- Improve the trust of AI predictions for users
- Improve the prediction performance of AI models
- Discover new domain knowledge



Adapted from Binder et al. 2018



Adapted from Hochuli et al. 2018



Adapted from Preuer et al. 2019

Samek et al. Explaining deep neural networks and beyond: A review of methods and applications.
Proceedings of the IEEE, 2021, 109(3), 247-278.



References



- Chapter 17 of Aurélien Géron' s book “Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow” (Ed. 2) , O’ Reilly, 2019. Chapter on VAE and GAN
- Jonathan Ho et al. “Denoising Diffusion Probabilistic Models” , NeurIPS 2020.
- Calvin Luo et al. “Understanding Diffusion Models: A Unified Perspective” , arXiv 2022.
- <https://cvpr2022-tutorial-diffusion-models.github.io/> (CVPR 2022 Tutorial on Denoising diffusion-based generative modeling)
- Shaoxiong Ji et al. “A Survey on Knowledge Graphs: Representation, Acquisition and Applications” , IEEE Transactions on Neural Networks and Learning Systems, 2021.
- Papers on Explainable AI have been listed at the bottom of the PPT slides

